

Additional Slides

Sentence-Level Factuality Classification in Brazilian Portuguese News

An NLP Classification Task Using Fine-Tuned Machine Learning Models

FACTNEWS DATASET

BRAZILIAN PORTUGUESE

TRANSFORMERS

António Santos

Paulo Silva

Gonçalo Dias

U. PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO



BERTimbau Tokenization Process Example:

Input: "A situação é gravíssima", afirmou o ministro.

True Label: Quote

Tokens: ["", 'A', 'situação', 'é', 'grav', '##íssima', "", ',', 'afirmou', 'o', 'ministro', '.']

Count: 12

Important observations: Tokenizes punctuation (relevant for quotes). Tokenizes affixes like “íssima” (relevant for bias)

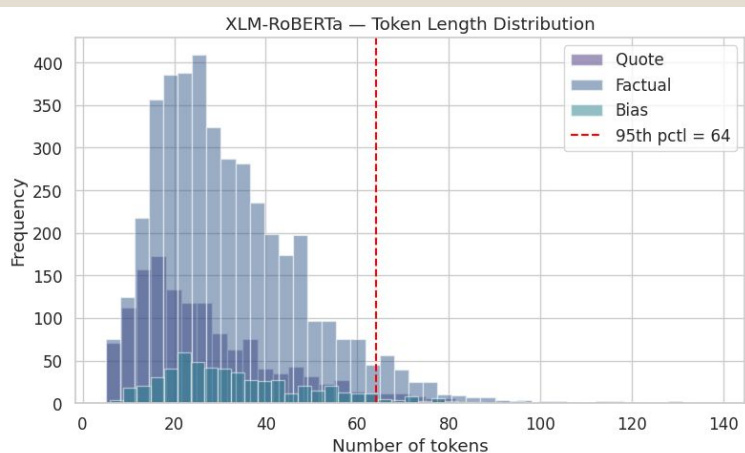
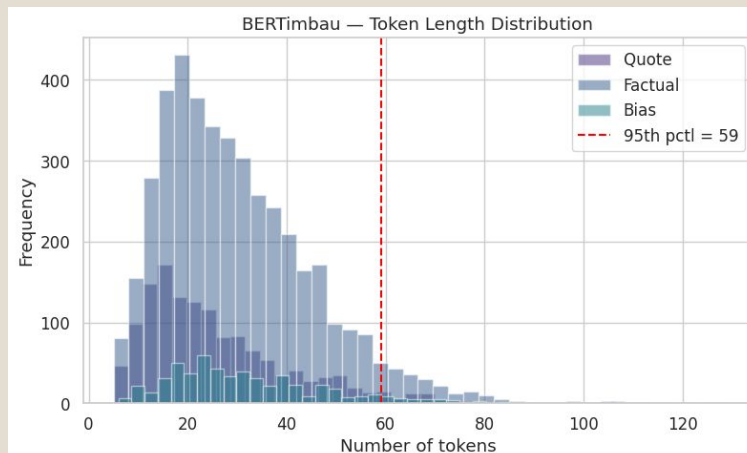
Example tokenized sample (BERTimbau):

Input: Milhares se aglomeram diante do Palácio de Buckingham para se despedir da rainha, que ocupou o trono...

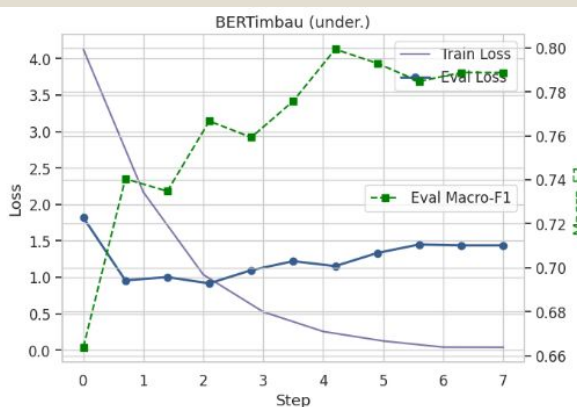
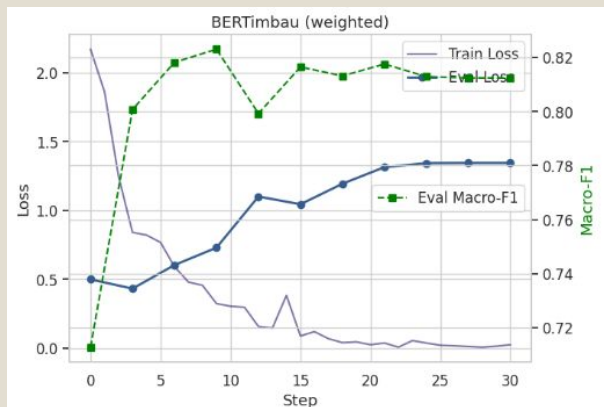
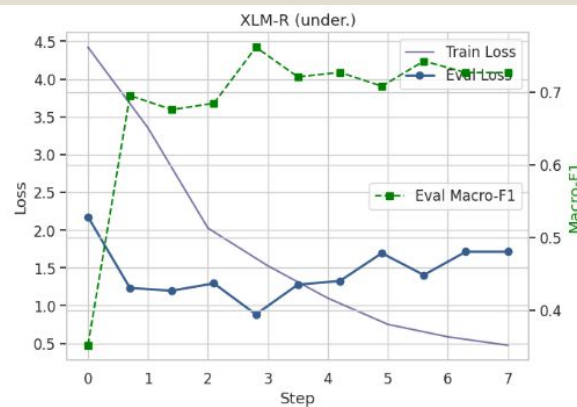
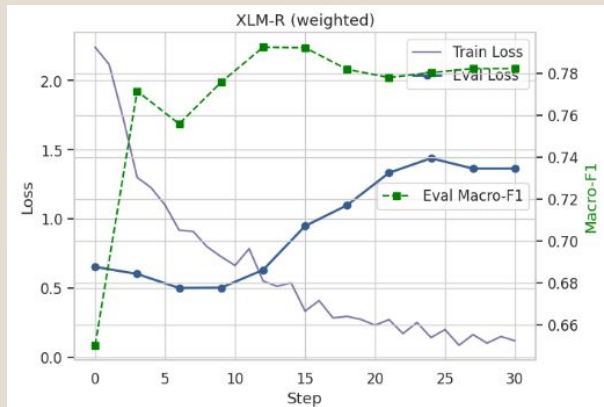
input_ids: [101, 2209, 5549, 176, 11621, 288, 4271, 171, 4976, 125]... (len=36)

attention_mask: [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]... (len=36)

label: 1 (Factual)



3-class Training Curves



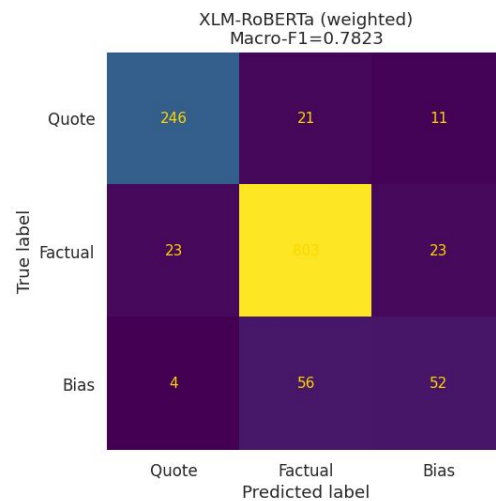
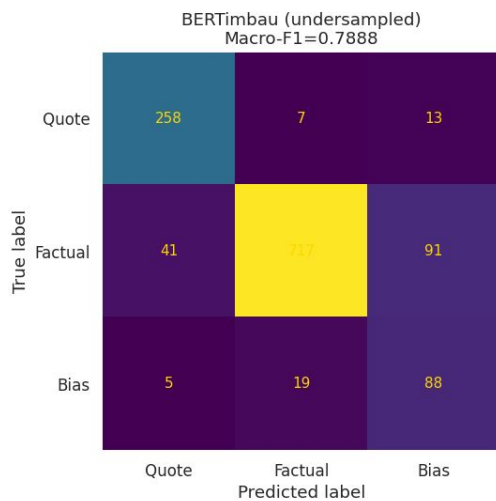
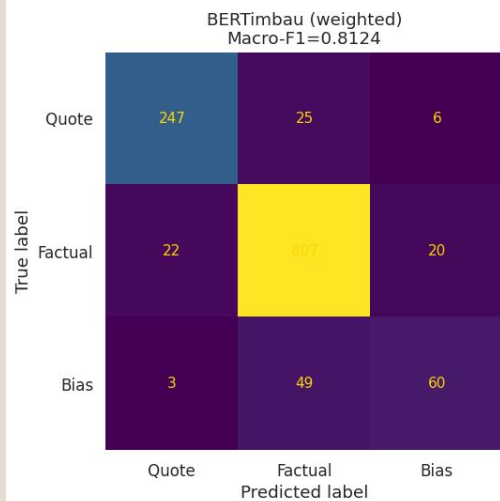
Full fine-tuning:
108,925,443 trainable
parameters (100%)

LoRA fine-tuning:
2,681,091 trainable
parameters (2.46%)

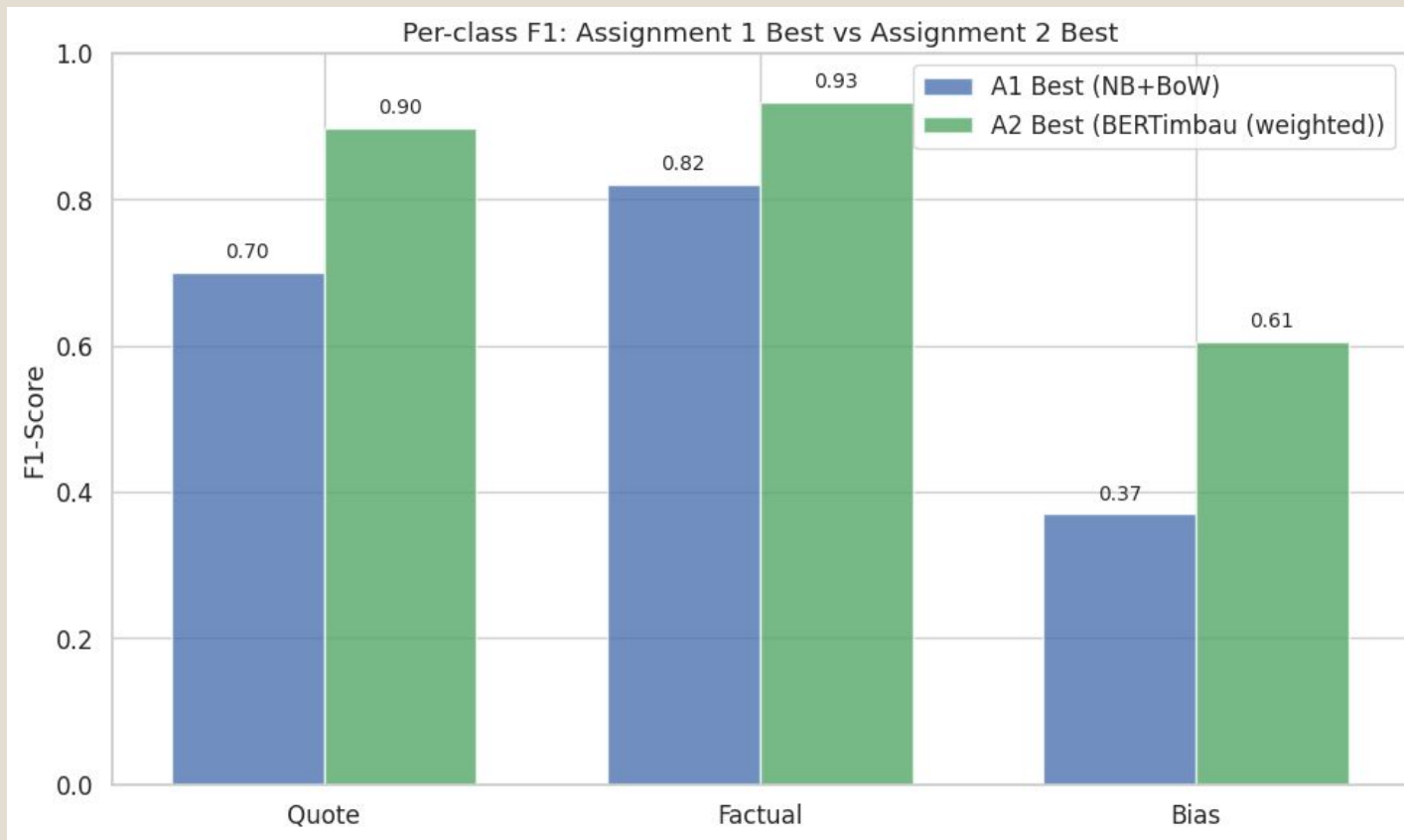
Reduction: 97.5%

Confusion Matrices: 3-class

Confusion Matrices — Transformer Models



Tuning Performance Comparison: A1 VS 3-Class Transformers

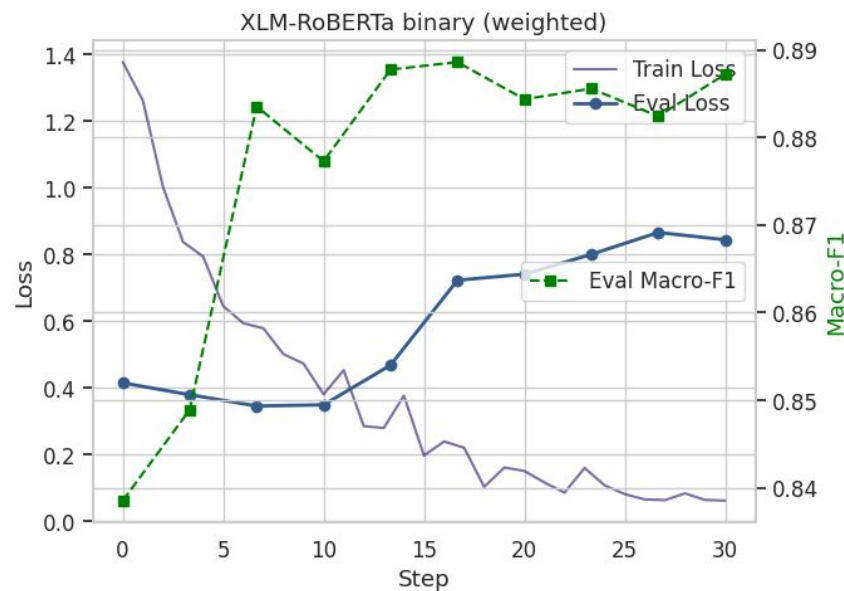
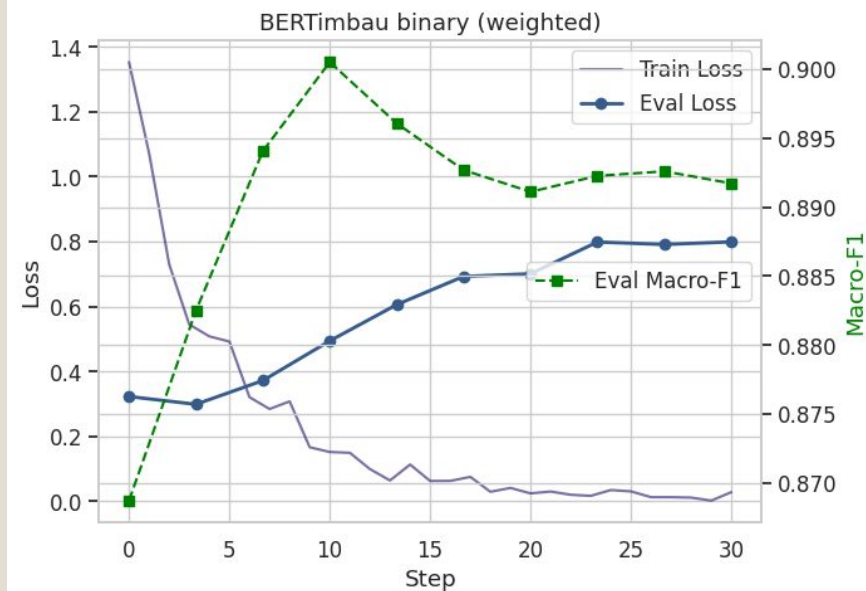


Binary Training Curves

Full fine-tuning:
108,924,674 trainable
parameters (100%)

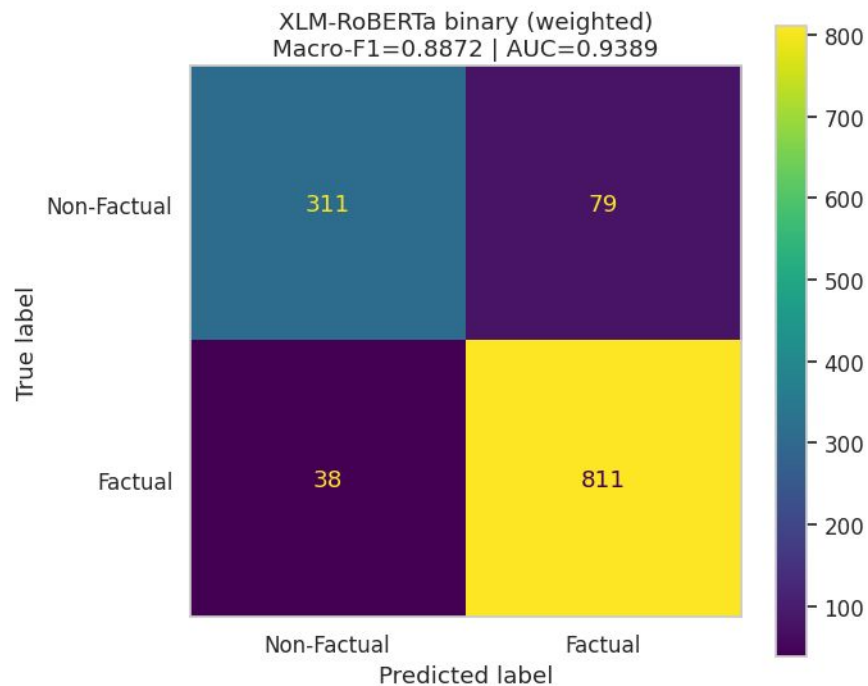
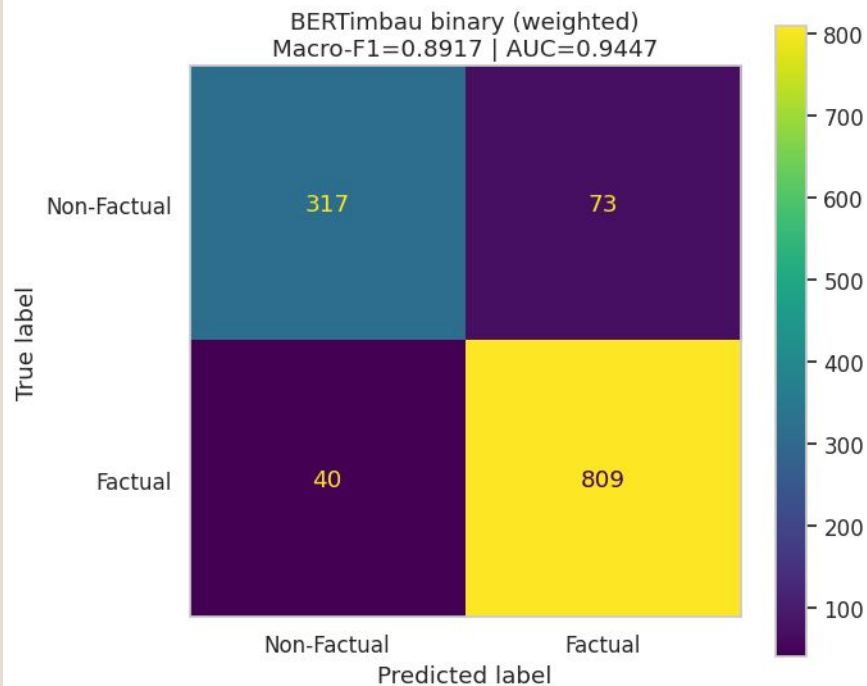
LoRA fine-tuning:
2,680,322 trainable
parameters (2.46%)

Reduction: 97.5%



Confusion Matrices: Binary

Confusion Matrices — Binary Classification (Non-Factual vs Factual)



Transformer Tuning Performance Comparison: 3-Class VS Binary

